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#University Category



**WORLD
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RANKINGS**

NO.1 PVT. UNIVERSITY IN
ACADEMIC REPUTATION IN INDIA



ACCREDITED GRADE 'A'
BY NAAC



PERFECT SCORE OF 150/150 AS A TESTAMENT
TO EXCEPTIONAL E-LEARNING METHODS

1



Applied Machine Learning

Unit 04 – Lecture 05: Data Cleaning

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Autumn 2026

Missing data, outliers and transformation

Topic focus: **the model is the easy part**

Repository: <https://github.com/tali7c/Applied-Machine-Learning>

Sources: Han, Kamber and Pei, *Data Mining: Concepts and Techniques* (3e), Ch. 3.; Müller and Guido, *Introduction to Machine Learning with Python*, Ch. 3–4.; James, Witten, Hastie, Tibshirani and Taylor, *An Introduction to Statistical Learning with Python*, Ch. 5–6, 12.; Scikit-learn documentation (preprocessing, model selection), accessed 2026-08-04.

Why clean
ooooo

Missing
oooooooooooooooo

Shape
ooooooo

Outliers
oooooooooooooooo

Transform
ooooooooooooo

Order
ooooo

Summary
oo

Quick Links

Why clean

Missing data

Outliers

Transformation

Order of operations

Summary

Agenda

Data preprocessing: why bother

Handling missing data

Describing the shape of a column

Handling outliers

Data transformation

The order of operations

Summary

How this lecture works

Every idea appears twice

1. **By hand** — eight marks, thirty-five sensor readings, nine prices. Small enough to check on paper.
2. **In code** — the same arithmetic on real data, so you see that `SimpleImputer` does exactly that and nothing more.

Commit before you look

Four slides ask you to predict a number before it is revealed. Write your guess down. Being wrong on paper is how the idea sticks.

The centrepiece is **a rule that cannot detect the outlier sitting in front of it**. Everything before it exists to make that failure obvious.

Where we are

Units I–III gave us

A model, a loss, and five ways to descend it. All of them assumed a clean numeric matrix X with no holes in it.

Real data is not like that

Values are missing, mistyped, duplicated, in the wrong units, or simply wrong. **No optimiser on the last four slide decks can rescue any of that.**

Unit IV starts here

Data preprocessing: cleaning (today), then feature engineering and reduction, then splitting and validation.

Today's four questions

What do I do about a hole? About a value that looks impossible? About a variable with a long tail? **And in what order?**

What “dirty” actually means

Defect	Looks like	Today?
incomplete	a blank cell, a NaN, a -1 that means “not asked”	yes
noisy	a reading that is wrong by a little	yes
outlying	a reading that is wrong by a lot — or right and rare	yes
skewed	one long tail, so a mean means nothing	yes
inconsistent	"M", "Male", "male", 1	L6
duplicated	the same patient entered twice	L6

Han, Kamber and Pei's four tasks

Cleaning (today), **integration**, **reduction** and **transformation** (today and L6). Cleaning is first because everything downstream inherits its mistakes.

Two datasets, one seed

The data

Two bundled tables: `load_diabetes` (442 patients, 10 features, used in both scaled and raw units) and `load_breast_cancer` (569 patients, 30 features). Nothing is downloaded.

The randomness

One seed for the whole lecture:
`np.random.default_rng(0)`,
`random_state=0`. Every listing that shuffles or samples shows it.

Two deliberate exceptions

The 35 sensor readings come from `default_rng(7)`: that seed *defines the data*, so it stays. The repeated-split experiments sweep `random_state` over `range(0, 20)` or `range(0, 300)`, because the spread across splits *is* the measurement. Run the code as written and every number here comes back.

Predict first #1

load_breast_cancer: 569 patients, 30 features. Knock out **5% of the individual cells**, completely at random. Then do the obvious thing — drop every row that has a hole in it.

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rows with at least one missing value	455 of 569
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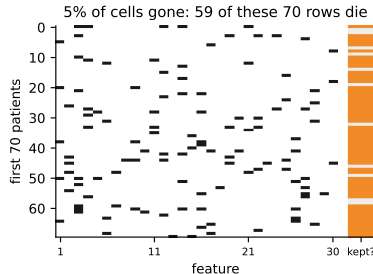
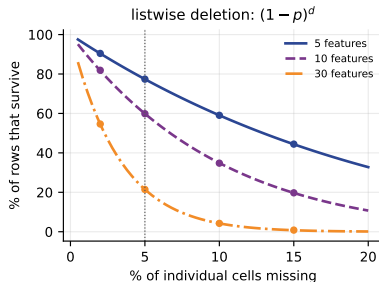
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Five per cent of the cells cost you eighty per cent of the rows

A row survives only if all 30 of its cells do: $0.95^{30} = 0.215$. **Listwise deletion is exponential in the number of columns**, and nobody's intuition is exponential.

Drawn: the exponent bites



Left: $(1 - p)^d$, measured against the formula. At $p = 5\%$ five columns keep 77% of rows and thirty columns keep 21%. Right: the actual mask. Black cells are missing; the orange strip is the verdict. **Fifty-nine of these seventy patients die of one missing number.**

Was deleting them at least safe?

accuracy on the 114 survivors	0.9735	+/- 0.0216
accuracy imputing instead, all 569 rows	0.9719	+/- 0.0129
baseline with no missingness at all	0.9807	+/- 0.0065

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Be honest about this

On *this* easy dataset deletion did not lose accuracy. It lost **455 patients** and widened the spread of the estimate by $1.7\times$.

It is not always this kind

On `load_diabetes` at 30% missing, listwise deletion leaves **9 training rows** for 10 features and the fitted model scores $R^2 = -4.70$.

$R^2 = 1 - SS_{\text{res}}/SS_{\text{tot}}$, so $R^2 < 0$ means *worse than always predicting $\bar{y}$* .

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Deletion is a decision to throw away evidence. Sometimes it is the right one. It is never the free one.

Three questions to ask about a hole

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The mistake this section exists to prevent

Reaching for `fillna(df.mean())` without asking any of the three.

By hand: eight marks, two missing

55, 62, ?, 71, 58, ?, 90, 64

Sort the six we have: 55, 58, 62, 64, 71, 90.

mean	$(55 + 62 + 71 + 58 + 90 + 64)/6 = 400/6$	66.6667
median	$(62 + 64)/2$	63.0000
sd ($n - 1$)	$\sqrt{\frac{1}{5} \sum (x_i - \bar{x})^2}$	12.6754

Mean, median or mode?

Mean for a symmetric numeric column. **Median** when it is skewed — here 90 drags the mean 3.7 marks above the median. **Mode** for a categorical column, where a mean is meaningless.

By hand: what mean imputation does to the spread

Fill both holes with $\bar{x}$. The mean cannot move — you added two copies of it. The *sum of squared deviations* cannot move either, because the two new deviations are zero. But you now divide it among more values:

$$\sigma_{\text{after}}^2 = \frac{\sum_{i=1}^m (x_i - \bar{x})^2}{n} = \frac{m}{n} \sigma_{\text{before}}^2 \implies \frac{\sigma_{\text{after}}}{\sigma_{\text{before}}} = \sqrt{\frac{m}{n}} = \sqrt{1 - p}.$$

Here $m = 6$ observed of $n = 8$, so the population variance is multiplied by $6/8 = 0.75$ and the sample variance by $(m - 1)/(n - 1) = 5/7 = 0.7143$.

This is not an approximation

It is exact, and it is the reason mean imputation is the worst defensible thing you can do. **You have invented confidence you do not have.**

The same thing in code

```
marks = np.array([55, 62, np.nan, 71, 58, np.nan, 90, 64])
obs = marks[~np.isnan(marks)]
for name, fill in (("mean", obs.mean()), ("median", np.median(obs))):
    f = np.where(np.isnan(marks), fill, marks)
    print(name, f.mean(), f.std(ddof=1))
```

```
mean of observed      66.6667
median of observed    63.0000
sd of observed (ddof=1) 12.6754
mean imputed: mean 66.6667 sd 10.7127 sd is now 84.5% of the truth
median imputed: mean 65.7500 sd 10.8463 sd is now 85.6% of the truth
```

$10.7127/12.6754 = 0.8452$, and $\sqrt{5/7} = 0.8452$. The formula and the library agree to four decimal places, because they are the same arithmetic.

Predict first #2

Take a real column — body mass index for the 442 diabetes patients, $sd = 4.42 \text{ kg/m}^2$.

Hide **40%** of it at random and fill every hole with the mean of what is left.

What is the standard deviation now? And what happens to its correlation with the target?

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```
p=0.1  measured sd ratio 0.9486  sqrt(1-p) = 0.9487
p=0.3  measured sd ratio 0.8363  sqrt(1-p) = 0.8367
p=0.5  measured sd ratio 0.7059  sqrt(1-p) = 0.7071
p=0.7  measured sd ratio 0.5447  sqrt(1-p) = 0.5477
complete data      corr(BMI, y) = 0.5865
40% mean-imputed   corr(BMI, y) = 0.4538 (77% of it survives)
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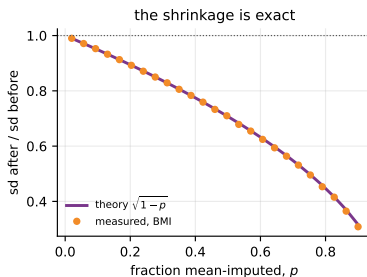
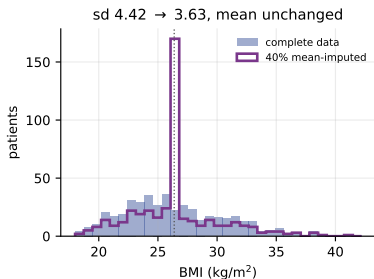
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The variance shrinks by $1 - p$, and a quarter of the signal goes

Four values of p , 2000 repetitions each, never more than 0.0031 from $\sqrt{1 - p}$. **Every correlation, every regression coefficient and every p -value computed afterwards is attenuated**, and nothing in the output warns you.

Drawn: a spike where a distribution used to be



Left: the same column before and after. Mean imputation does not blur the distribution — it **piles every hidden patient onto a single value**, 4.4181 → 3.6309. Right: the average of 400 draws per point against $\sqrt{1-p}$, largest disagreement 0.0080 over the whole range.

MCAR, MAR, MNAR: the taxonomy that matters

Write R for the indicator “this value is missing”, X_{obs} for what you can see and X_{mis} for what you cannot.

MCAR missing completely at random **MAR** missing at random **MNAR** missing not at random

	Depends on	Example
MCAR	nothing: $P(R)$ constant	a tube was dropped in the lab
MAR	X_{obs} only	BMI recorded only for the high-blood-pressure patients
MNAR	X_{mis} itself	the heaviest patients decline to be weighed

Why you should care

Under **MCAR** the complete cases are a fair sample — you lose power, not correctness. Under **MAR** a good imputer that uses the other columns can recover the truth. Under **MNAR** *nothing in the data can*, because the information you need was never recorded.

The three mechanisms, measured

Same column, same 40–48% missing, three different reasons:

```
rng = np.random.default_rng(0) # the one seed for this lecture
mcar = rng.random(n) < 0.40 # a coin
mar = bp < np.median(bp) # depends on BP, which we can see
mnar = bmi > np.quantile(bmi, 0.60) # depends on BMI itself
```

complete-data mean BMI	26.3758 kg/m2	(n = 442)	
MCAR	35.3% missing	complete-case mean	26.4913 bias +0.1155
MAR	48.0% missing	complete-case mean	27.7230 bias +1.3473
MNAR	40.0% missing	complete-case mean	23.4253 bias -2.9505

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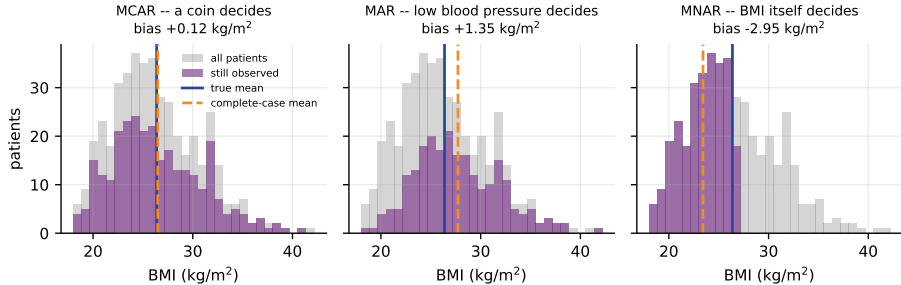
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Throwing away the incomplete rows is *almost* free under MCAR (+0.12, and it would change sign with another draw), wrong by +1.35 under MAR, and wrong by −2.95 under MNAR — **two thirds of a standard deviation**, from a decision you never made.

Drawn: the same histogram, three ways of losing it



Grey is everyone; purple is who is left. Under MCAR the purple bars are a shrunken copy of the grey. Under MNAR the right-hand tail is **simply gone**, and no amount of cleverness puts it back.

Can a better imputer repair it?

IterativeImputer (MICE) predicts each column from all the others, round-robin, until it settles.

MCAR mean	imputed mean	26.4913	bias	+0.1155
MCAR MICE	imputed mean	26.4763	bias	+0.1005
MAR mean	imputed mean	27.7230	bias	+1.3473
MAR MICE	imputed mean	26.2170	bias	-0.1588
MNAR mean	imputed mean	23.4253	bias	-2.9505
MNAR MICE	imputed mean	23.7924	bias	-2.5834

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Read the middle pair, then the last pair

Under MAR, MICE cuts the bias from +1.35 to -0.16 — an 8.5× **improvement**, because blood pressure was recorded and BMI can be predicted from it. Under MNAR it moves -2.95 to -2.58 and stops. **The taxonomy is not academic: it tells you whether to bother.**

And what it does to a coefficient you would report

Regress the target on all ten features and read off the BMI coefficient.

true BMI coefficient	5.603
MNAR + mean imputation	-0.413
MNAR + MICE	2.299

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Regress the target on all ten features and read off the BMI coefficient.

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The sign flipped

Mean-imputing an MNAR column does not merely weaken the relationship. It replaces the entire upper 40% of the distribution with one constant, and the fitted slope through what remains points the *other way*. **You would have written “BMI is unrelated to disease progression” in a report.**

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Prediction is more forgiving than inference — but only because nobody reads a predictor's coefficients.

So which imputer should you use?

load_diabetes, holes punched in the *training* matrix only, clean test set, 20 random splits:

	10% missing	30% missing
oracle, no missingness	0.4762	0.4762
drop incomplete rows	0.4264	-4.6952 (+/- 7.88)
mean	0.4721	0.4351
median	0.4727	0.4364
constant 0	0.4723	0.4365
KNN, k=5	0.4743	0.4591
iterative (MICE)	0.4751	0.2639 (+/- 0.50)

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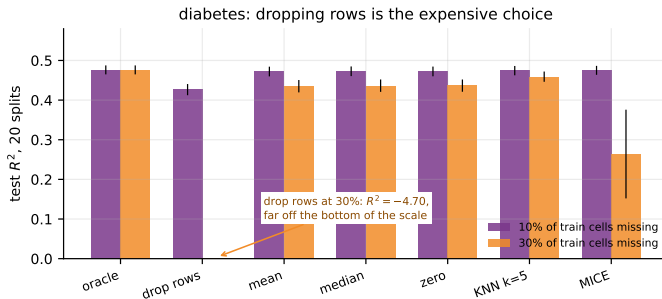
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Deletion costs 0.05; the choice of imputer costs 0.003

At 10% every imputer lands within 0.003 of every other and within 0.004 of the oracle. **Do not agonise over the imputer. Do not drop the rows.** At 30%, KNN pulls ahead — and MICE became unstable on two of twenty splits, which is why its error bar is a factor of ten larger.

Drawn: the same table



Purple 10%, orange 30%; error bars are standard errors over 20 splits. Six of the seven bars are the same height. The one that is not is the one everybody's first instinct reaches for.

Missingness is itself data

A weak feature, missing 80% of the time for one class and 20% for the other — a test ordered only when the clinician already suspects something.

```
SimpleImputer(strategy="mean", add_indicator=True)
```

```
mean imputation only      0.6294
mean imputation + indicator 0.7969
difference                 +0.1675
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Sixteen points of accuracy

from a column of True/False that you get for free with one keyword argument.

And a warning

If the reason for the hole will not exist at prediction time, that indicator is **leakage**, not a feature.

Two numbers are not enough

A mean and a standard deviation are the first two moments. They say nothing about **shape** — two columns can match on both and look nothing alike.

$$m_k = \frac{1}{n} \sum_i (x_i - \bar{x})^k \quad \Rightarrow \quad \underbrace{g_1 = \frac{m_3}{m_2^{3/2}}}_{\text{skewness}} \quad \underbrace{b_2 = \frac{m_4}{m_2^2}}_{\text{kurtosis}}$$

Skewness — an *odd* power

Sign survives, so one long tail fails to cancel.
 $g_1 > 0$ tail to the right, $g_1 < 0$ to the left, 0 balanced.

Kurtosis — an *even* power

Sign destroyed, magnitude counts. A deviation of 4 weighs $256 \times$ one of 1: a statement about **tails**.

$$\text{Excess kurtosis} = b_2 - 3$$

A normal has $b_2 = 3$ exactly, so subtracting puts the normal at zero. `scipy` returns the **excess** form by default; many textbooks report b_2 . Say which you mean.

By hand: five numbers

$x = (1, 2, 2, 3, 7)$, so $\bar{x} = 3$ and the deviations are $(-2, -1, -1, 0, 4)$.

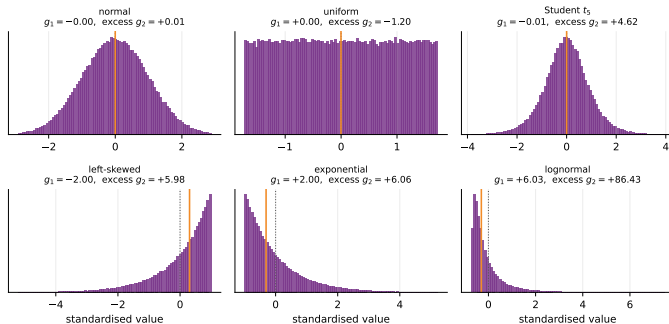
m_2	$(4 + 1 + 1 + 0 + 16)/5$	4.4
m_3	$(-8 - 1 - 1 + 0 + 64)/5$	10.8
m_4	$(16 + 1 + 1 + 0 + 256)/5$	54.8

In m_4 the single value 7 contributes 256 of the 274 — **93% from one point in five.**

```
skewness g1 = m3 / m2^1.5 = 10.8000 / 9.2295 = +1.1702
kurtosis b2 = m4 / m2^2   = 54.8000 / 19.3600 = +2.8306
excess      = b2 - 3      = -0.1694
```

Positive skew, as the picture demands — but the excess kurtosis is **negative**, next to an obvious outlier. Hold that thought.

Drawn: six shapes, two numbers each



The uniform is the one to remember: perfectly symmetric, $g_1 = 0$, yet excess $g_2 = -1.2$ because it stops dead and has no tails at all. Negative is not a defect — it is a bounded variable.

Predict first: what does kurtosis measure?

A normal, and a mixture: 95% drawn with sd 0.8, 5% with sd 3.0. Both standardised to sd 1. Textbooks say kurtosis is “peakedness”.

Is the mixture's high kurtosis coming from its peak or its tails?

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	density at 0	% $ z >3$	excess kurt	share of sum d^4
normal	0.4002	0.2830	0.0079	0.1134
95/5 mixture	0.4941	1.5005	9.0433	0.8761

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Both are true — and the decomposition settles it

The mixture *is* 24% taller at the centre. But split $\sum d^4$ by where it comes from: **1.5% of the rows supply 88% of the sum**. Small deviations to the fourth power are nothing. Kurtosis is an outlier-frequency statistic.

Both estimates are noisy, and the thresholds are folklore

" $|g_1| > 0.5$ means skewed." Draw 1000 samples from a *perfectly symmetric* normal and look at what they report:

n	mean skew	sd skew	sqrt(6/n)	sd exc k
20	-0.0387	0.4567	0.5477	0.7289
30	-0.0223	0.3932	0.4472	0.6891
100	-0.0037	0.2394	0.2449	0.4468
1000	+0.0032	0.0787	0.0775	0.1531

at $n=30$, 20.4% of samples from a perfectly symmetric distribution report $|\text{skewness}| > 0.5$, the threshold often quoted as 'skewed'

$SE \approx \sqrt{6/n}$ for skewness, $\sqrt{24/n}$ for excess kurtosis. At $n = 30$, **one sample in five** clears a threshold it should never reach.

And both have a ceiling, exactly like the z-score

$$|g_1| \leq \frac{n-2}{\sqrt{n-1}}, \quad b_2 \leq \frac{n^2 - 3n + 3}{n-1}$$

n	one outlier	best 2-point split	$(n^2-3n+3)/(n-1) - 3$
5	0.2500	0.2500	0.2500
30	25.0345	25.0345	25.0345
569	564.0018	564.0018	564.0018

That is why five points gave -0.17

At $n = 5$ the largest excess kurtosis *any* five numbers can produce is **0.25**. Our example was most of the way to the maximum. It never said “light tails”; it said “as heavy as five points are permitted to be”. Below about fifty rows, plot the column instead.

Why it decides the next two sections

column	skew	exc k	mean	median	gap %
mean texture	+0.649	+0.74	19.290	18.840	2.4%
worst area	+1.854	+4.35	880.583	686.500	28.3%
area error	+5.433	+48.77	40.337	24.530	64.4%

- **Which imputer.** At $g_1 = 0.65$ mean and median differ by 2.4% and the choice hardly matters. At $g_1 = 5.43$ the mean sits 64% above the median and above 71% of the rows.
- **How to read an outlier rule.** $g_2 = 48.8$ guarantees a lot of flagged rows — and they are the shape, not contamination.
- **Whether a transform worked.** Both numbers give you a target.

Three kinds of outlier, one word

An error

Age = 200. A stuck sensor. A pressure in the wrong units.

Fix or remove it.

A rare truth

A genuine very large tumour.

Keep it. Use a robust method.

The point

Fraud. A failing machine.
A new phenomenon.

It is the signal.

No statistical rule can tell these three apart

A rule tells you a point is *unusual*. Only domain knowledge tells you what to do about it. **Detection is statistics; disposal is a judgement you must be able to defend.**

Iglewicz and Hoaglin's three words: *labelling, accommodation, identification*. Today is mostly labelling.

Three rules you will meet

	Statistic	Flag when	Breakdown point
z-score	$z_i = \frac{x_i - \bar{x}}{s}$	$ z_i > 3$	0%: one point moves both $\bar{x}$ and s
Tukey / IQR	$IQR = Q_3 - Q_1$	outside $Q_{1,3} \mp 1.5 IQR$	25%
modified z	$M_i = \frac{0.6745 (x_i - \tilde{x})}{MAD}$	$ M_i > 3.5$	50%: median and MAD

Where the two magic constants come from

$MAD = \text{median } |x_i - \tilde{x}|$. For Normal data $MAD \approx 0.6745 \sigma$, so dividing by 0.6745 puts M back on the z scale. The 1.5 in Tukey's fence flags 0.70% of Normal data; $|z| > 3$ flags 0.27%.

By hand: thirty-five readings from one sensor

Thirty genuine readings scattered around 10.0, then the sensor sticks and reports 100.0 five times.

mean $\bar{x}$	22.503	median $\tilde{x}$	9.950
sd s	32.109	MAD	0.540
Q_1	9.245	Q_3	10.330

Now score one stuck reading, $x = 100$, with each rule:

$$z = \frac{100 - 22.503}{32.109} = 2.4135, \quad M = \frac{0.6745(100 - 9.950)}{0.540} = 112.48.$$

2.41 against a threshold of 3

The five stuck readings dragged $\bar{x}$ from 9.59 up to 22.50 and inflated s from 0.84 to 32.11. **They are being measured against a ruler they themselves bent.**

Predict first #3

Thirty good readings near 10. Five readings of exactly 100 — as wrong as a reading can be, and you can see them from across the room.

How many of the five does the $|z| > 3$ rule catch? What about the IQR fence and the modified z ?

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How many of the five does the $|z| > 3$ rule catch? What about the IQR fence and the modified z ?

rule	flags	of which stuck	of which genuine
$ z > 3$	0	0	0
outside 1.5*IQR fence	6	5	1
modified z > 3.5	5	5	0

sd WITHOUT the five stuck readings 0.837
z of a stuck reading would then be 108.1

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Zero. The most-taught rule in the subject finds none of them

And it is not close: 2.41 against 3. Remove the outliers first and the very same reading scores $z = 108$.

This is masking: an outlier inflates the standard deviation it is being divided by, and hides inside its own damage.

The same thing in code

```
z = (v - v.mean()) / v.std(ddof=1)
q1, q3 = np.percentile(v, [25, 75]); iqr = q3 - q1
mad = np.median(np.abs(v - np.median(v)))
mz = 0.6745 * (v - np.median(v)) / mad
```

```
mean 22.503   sd 32.109   median 9.950   MAD 0.540
Q1 9.245   Q3 10.330   IQR 1.085   upper fence Q3+1.5*IQR 11.957
a stuck reading: z = 2.4135   (needs > 3)   modified z = 112.48   (needs > 3.5)
```

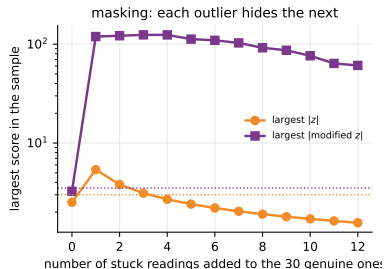
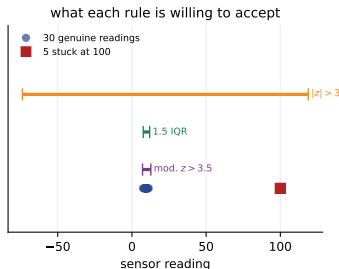
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a stuck reading: z = 2.4135   (needs > 3)   modified z = 112.48   (needs > 3.5)
```

Four lines each. The whole difference is that the middle rule uses **quartiles** and the third uses **a median and a median** — and a median does not care how far away the far-away points are.

Drawn: masking, and why it gets worse



Left: what each rule is willing to accept. The z interval reaches from -74 to $+119$; the other two are slivers around 10. Right: add stuck readings one at a time. **One outlier is caught at $z = 5.38$, two at 3.81 , three at 3.11 , and from the fourth onwards none ever again.** The modified z stays above 60 throughout.

By hand: the rule has a ceiling

For a sample of n values with sample standard deviation s (the $n - 1$ one), no observation can have

$$|z| > \frac{n-1}{\sqrt{n}}.$$

n	largest possible $ z $	can the rule $ z > 3$ ever fire?
3	1.1547	NO -- not ever
5	1.7889	NO -- not ever
10	2.8460	NO -- not ever
11	3.0151	yes
50	6.9296	yes

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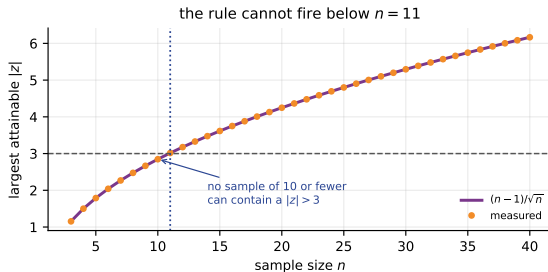
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Below $n = 11$ the rule is not strict — it is inert

Whatever the data, whatever the contamination. If you have ten measurements and you apply $|z| > 3$, you have written a line of code that can only ever return “no outliers”. NIST states the bound in exactly these words, and the fix in the next line: use the modified z .

Drawn: the ceiling



Theory and measurement agree to 9×10^{-16} . The curve crosses 3 between $n = 10$ (2.846) and $n = 11$ (3.015). **The threshold that everybody memorises has a domain of validity that nobody mentions.**

The opposite failure: a skewed column

The most skewed column of `load_breast_cancer` — 'area error', skewness 5.43, no contamination at all.

rule	flags	% of rows
<code> z > 3</code>	6	1.1%
1.5*IQR fences	65	11.4%
<code> modified z > 3.5</code>	81	14.2%

under a perfect Normal these rules flag 0.27%, 0.70% and 0.35% of rows

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now Yeo-Johnson the column first (skewness 5.433 -> 0.069) and repeat

z > 3	0	0.0%
1.5*IQR fences	1	0.2%
modified z > 3.5	0	0.0%

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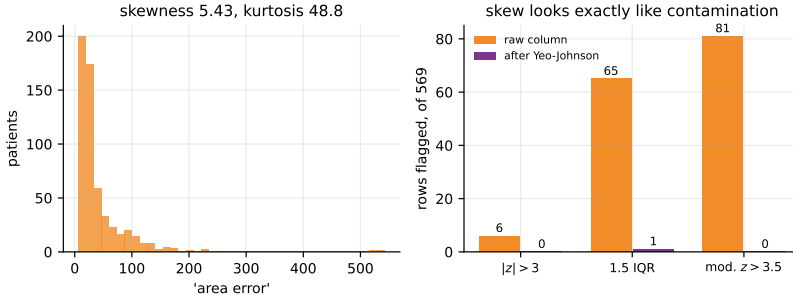
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Eighty-one “outliers” that were never outliers

Every one of these rules assumes symmetry. On a long right tail they flag the tail. **Transform first, then look for outliers** — and that is the whole argument for the next section.

Drawn: skew looks exactly like contamination



Left: the raw column, kurtosis 48.8. Right: how many rows each rule condemns before and after a single power transform. $6 \rightarrow 0$, $65 \rightarrow 1$, $81 \rightarrow 0$ — and not one data value was deleted.

Numerical methods, 1: binning and smoothing

Nine sorted prices, three equal-depth bins. This is Han, Kamber and Pei's standard treatment of *noisy* data, and it is a favourite exam question.

Bin	Values	mean	median	boundaries
1	4, 8, 15	9	8	4 and 15
2	21, 21, 24	22	21	21 and 24
3	25, 28, 34	29	28	25 and 34

by bin means	9, 9, 9, 22, 22, 22, 29, 29, 29
by bin medians	8, 8, 8, 21, 21, 21, 28, 28, 28
by bin boundaries	4, 4, 15, 21, 21, 24, 25, 25, 34

Read the third row

Each value is replaced by whichever *end* of its bin is closer: 8 is nearer 4 than 15, so it becomes 4; 15 stays 15.

Numerical methods, 2: five things to do to an outlier

Back to the thirty-five sensor readings. The truth is mean 9.586, sd 0.837.

leave them alone	mean	22.503	sd	32.109	n	35
winsorise at 5 and 95	mean	22.523	sd	32.100	n	35
winsorise at 15 and 85	mean	9.903	sd	0.839	n	35
cap at the IQR fence	mean	9.925	sd	1.143	n	35
drop by modified z	mean	9.586	sd	0.837	n	30
the 30 genuine readings	mean	9.586	sd	0.837	n	30

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Look at the second row before anything else

Winsorising at the 5th and 95th percentiles did **nothing**, because $5/35 = 14\%$ of the data is contaminated and you told it to clip 10%. **A fixed percentile is a guess about how much dirt there is.** The IQR fence and the modified z work it out from the data.

And the scaler you choose is an outlier decision

```
StandardScaler().fit_transform(sensor.reshape(-1, 1)) # (x - mean) / sd  
RobustScaler().fit_transform(sensor.reshape(-1, 1)) # (x - median) / IQR
```

```
StandardScaler  30 genuine readings span -0.4747 to -0.3527 (width 0.1220)  
RobustScaler    30 genuine readings span -2.2765 to  1.2811 (width 3.5576)
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```

The same thirty readings, squashed into 0.12 of a unit or spread over 3.56.
StandardScaler spent its whole dynamic range accommodating five wrong numbers, and every distance-based model downstream now thinks those thirty patients are identical.

Why transform a variable at all

Four honest reasons

1. Symmetry: a mean and an sd stop lying.
2. Linearity: a curved relationship straightens.
3. Equal spread: residual variance stops growing with the fit.
4. Outlier rules start working (previous section).

The ladder of powers

..., $-1/x$, $-1/\sqrt{x}$, $\log x$, $\sqrt{x}$, x , x^2 , ...

Move **down** to pull in a right tail, **up** for a left one. $\log$ sits in the middle and is the one you will use most.

Two things a transform is not

It is not scaling (that is L6, and it does not change shape). And it is not free: after $\log$, a coefficient is a statement about *percentage* change, and you must say so.

Four transformations, one real column

'area error': 569 values, minimum 6.80, maximum 542.2.

transformation	skewness	kurtosis
none	5.4328	48.7672
sqrt(x)	2.1363	7.8193
log(x)	0.7955	0.3622
1/sqrt(x)	0.0447	-0.3685
Box-Cox, lam=-0.436	0.0584	-0.4094
Yeo-Johnson	0.0691	-0.4561

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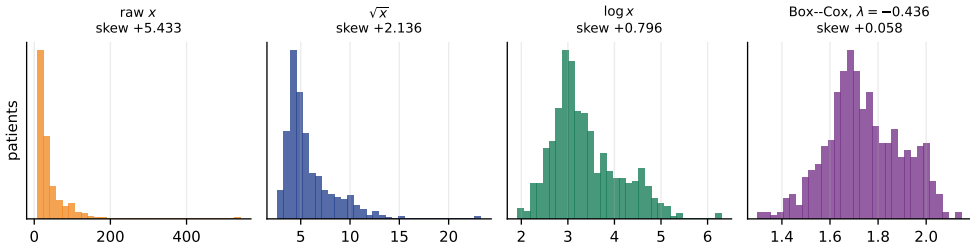
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The ladder is discrete, and the right rung is between two of them

That is precisely the gap Box-Cox fills.

Drawn: the same 569 numbers, four ways



Nothing has been deleted and no value has changed its rank. All that changed is the ruler the values are printed on — and the fourth panel is a variable you can put a mean and a standard deviation on.

Box–Cox and Yeo–Johnson

$$x^{(\lambda)} = \begin{cases} \frac{x^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \log x, & \lambda = 0 \end{cases} \quad x > 0 \text{ required}$$

Why the -1 and the $/\lambda$

So that the family is *continuous* at $\lambda = 0$: $(x^\lambda - 1)/\lambda \rightarrow \log x$. Without them, $\lambda \rightarrow 0$ would give the constant 1.

Yeo–Johnson

The same idea rewritten so that it works for $x \leq 0$: it applies the Box–Cox form to $x + 1$ on the positive side and a mirrored version to $-x + 1$ on the negative side.

λ is chosen by **maximum likelihood** under a Normal model, not by minimising skewness — which is why the two answers differ slightly.

By hand: the family on four numbers

$x = (1, 4, 16, 64)$ — a *geometric* sequence, each value $4 \times$ the last.

```
lambda +1.00 -> [ 0.  3. 15. 63.]  
lambda +0.50 -> [ 0.  2.  6. 14.]  
lambda +0.00 -> [0.    1.3863 2.7726 4.1589]  
lambda -0.50 -> [-0.    1.    1.5   1.75]  
lambda -1.00 -> [-0.    0.75  0.9375 0.9844]
```

Check one with a pen: $\lambda = -0.5$, $x = 16$. $16^{-0.5} = 1/4 = 0.25$, so $(0.25 - 1)/(-0.5) = \mathbf{1.5}$ — the third entry.

Read the $\lambda = 0$ row

0, 1.3863, 2.7726, 4.1589 — the gaps are all $\log 4$. The log turned a **geometric** sequence into an **arithmetic** one. That is what “the log is the right ruler for multiplicative data” actually means.

Yeo–Johnson: the same idea on the whole real line

$$y^{(\lambda)} = \begin{cases} \frac{(y+1)^\lambda - 1}{\lambda}, & \lambda \neq 0, y \geq 0 \\ \log(y+1), & \lambda = 0, y \geq 0 \\ -\frac{(-y+1)^{2-\lambda} - 1}{2-\lambda}, & \lambda \neq 2, y < 0 \\ -\log(-y+1), & \lambda = 2, y < 0 \end{cases}$$

Four cases, one idea: the lower pair is the upper pair **reflected** — negate the input, swap λ for $2 - \lambda$, negate the result.

```
lambda 0.0 -> [-7.5   -1.5   -0.625  0.      0.4055  0.6931  1.3863]
lambda 1.0 -> [-3.    -1.    -0.5    0.     0.5    1.     3.   ]
lambda 2.0 -> [-1.3863 -0.6931 -0.4055  0.      0.625   1.5    7.5   ]

our implementation vs sklearn on 'area error', lambda = -0.483636
max |difference| = 0.000e+00
```

$\lambda = 1$ is the identity, every λ maps $0 \rightarrow 0$, and the $\lambda = 0$ and $\lambda = 2$ rows are mirror images.

Predict first: what picks λ ?

Maximise the Normal log-likelihood of the *transformed* column:

$$\ell(\lambda) = -\frac{n}{2} \log \hat{\sigma}_\lambda^2 + \underbrace{(\lambda - 1) \sum_i \log x_i}_{\text{Jacobian}}$$

Every summary drops the second term. What happens if you do?

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```
with the Jacobian term : lambda* = -0.4360
scipy.stats.boxcox      : lambda* = -0.4357
without the Jacobian term : lambda* = -2.0000 <-- runs to the edge

lambda -2.000    var 2.089073e-06
lambda -0.436    var 2.458963e-02
```

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```

The first term alone rewards crushing the data to a point

Smaller variance always scores better, so the optimiser runs to the edge of the grid. The Jacobian is the change-of-variables factor, and it charges for that squashing at exactly the right rate.

Two details that confuse everybody

```
standardize=False  lambda -0.4836  mean  +1.6509  sd  0.1258  skew +0.0691
standardize=True   lambda -0.4836  mean  -0.0000  sd  1.0000  skew +0.0691

lambda fitted on train only : -0.5527
lambda fitted on all rows   : -0.4836  (this one has seen the test set)
lambda fitted on test only  : -0.3386
```

standardize=True is the default

It transforms *and* z-scores. Skewness is identical either way — standardising is linear and cannot change shape.

λ is a *learned parameter*

Three splits, three values. Fit it on all the rows and you have leaked, exactly as with a scaler or an imputer. It goes in the Pipeline.

λ chooses a rung of the ladder

```
lambda -1.00  1/x (reciprocal)  skewness -0.8635
lambda -0.50  1/sqrt(x)         skewness -0.0447
lambda +0.00  log(x)            skewness +0.7955
lambda +0.50  sqrt(x)           skewness +2.1363
lambda +1.00  x (no change)     skewness +5.4328
maximum-likelihood lambda from scipy -0.4357
skewness there                               +0.0584
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```
a column that goes negative cannot use Box-Cox
diabetes 'bp': min -0.1124 max 0.1320 skewness +0.2897
stats.boxcox(bp) -> ValueError: Data must be positive.
Yeo-Johnson handles it: skewness +0.2897 -> +0.0226
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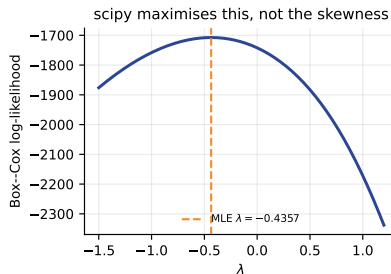
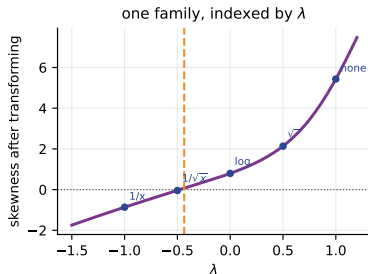
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Every rung you know is a value of λ , and λ is allowed to be -0.4357 .

Drawn: what scipy is actually maximising



Left: skewness against λ , with the five named rungs marked. Right: the Box-Cox log-likelihood, whose peak is at $\lambda = -0.4357$. The λ that would zero the skewness exactly is -0.4688 — close, but they are answers to different questions.

Does it help the model? It depends which model

20 random splits of load_breast_cancer:

GaussianNB	scale only	0.9304	Yeo-Johnson first	0.9462	+0.0157	(16/20)
LogisticRegression	scale only	0.9762	Yeo-Johnson first	0.9766	+0.0003	(5/20)

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Naive Bayes gains 1.6 points

It *assumes* each feature is Normal within a class. Making that assumption true is worth real accuracy, on 16 of the 20 splits.

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It makes no distributional assumption about X at all. +0.0003 is **noise**, and it **won 5 splits of 20**.

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Logistic regression gains nothing

It makes no distributional assumption about X at all. +0.0003 is **noise**, and it **won 5 splits of 20**.

Transform because the model assumes it, or because you need to read the variable — not as a ritual.

The single rule that governs all of it

Every number used to clean the data must be computed from the **training rows only**.

Step	The number it learns
imputation	the column mean, median, mode, or a fitted model
outlier removal	a threshold, or <i>which rows to keep</i>
transformation	λ
scaling	the mean and the standard deviation

Break it and your test score is a measurement of nothing

The score you report is no longer an estimate of performance on unseen data, because the data was not unseen. **How much you overstate by depends entirely on which step you got wrong** — and the range is enormous.

Predict first #4

The most human thing in data science: fit a model, look at the rows it gets badly wrong, decide they are outliers, delete them, and start again.

Concretely: fit on all 442 diabetes patients, delete the 10% with the largest residual, *then* split into train and test and report R^2 .

By how much does that overstate the honest number?

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LEAK C -- drop the rows your model cannot fit, THEN split. 200 splits
drop worst 5% leaky 0.5850 honest 0.4730 gap +0.1120 t +19.2 win 181/200
drop worst 10% leaky 0.6390 honest 0.4688 gap +0.1702 t +29.9 win 192/200
drop worst 20% leaky 0.7022 honest 0.4732 gap +0.2291 t +45.0 win 200/200
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0.47 became 0.64, on 192 splits out of 200

The honest column barely moves — cleaning the *training* set is fine. What moved is the test set: you removed exactly the patients your model was going to fail on. **You did not improve the model. You marked your own exam paper.**

Three sins, all measured, and they are not equal

```
LEAK A -- fit the scaler on everything, then split. 300 splits each
  n_train = 20 leaky 0.9338 honest 0.9304 gap +0.0034 t +6.0 win 172/300
  n_train = 60 leaky 0.9587 honest 0.9577 gap +0.0011 t +4.3 win 146/300
  n_train = 427 leaky 0.9774 honest 0.9772 gap +0.0002 t +1.7 win 18/300
LEAK B -- fit the imputer on everything. 30% missing, 60 splits
  mean leaky 0.9586 honest 0.9583 gap +0.0003 t +0.8 win 9/60
  KNN k=5 leaky 0.9535 honest 0.9552 gap -0.0017 t -1.4 win 18/60
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Be honest about the size of each one

Leak A is **real** — $t = 6.0$ over 300 splits is not noise — and it is **tiny**, 0.3 of a percentage point, shrinking to nothing as the training set grows. Leak B did not show a gap at all here. Leak C was worth 0.17 of R^2 .

Three sins, all measured, and they are not equal

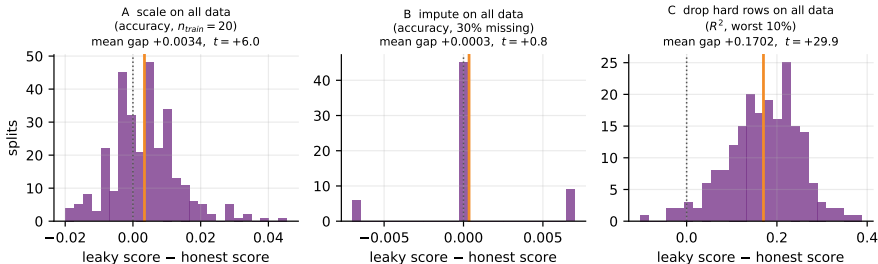
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LEAK A -- fit the scaler on everything, then split. 300 splits each
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The size of a leak is the amount of test information the step absorbs. A mean absorbs almost none. A decision about which rows to keep absorbs a great deal.

Drawn: the three gaps on the same axis idea



Each histogram is the per-split difference, leaky minus honest. Note the horizontal scales: 0.04, 0.008 and 0.4. **Do not learn “all leakage is catastrophic” and do not learn “leakage is overblown”.** Learn to ask what the step learned from the test set.

The pipeline that cannot leak

```
pipe = Pipeline([
    ("impute", SimpleImputer(strategy="median", add_indicator=True)),
    ("power", PowerTransformer(method="yeo-johnson")),
    ("scale", StandardScaler()),
    ("model", LogisticRegression(max_iter=5000)),
])
cross_val_score(pipe, X_with_holes, y, cv=5).mean()
```

```
one Pipeline, 5-fold cross-validation, 10% of cells missing
steps: impute -> power -> scale -> model
cross_val_score 0.9578 +/- 0.0140
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Every step is refitted inside every fold. The median, λ , the mean and the sd are all recomputed nine times, from four fifths of the data, and the held-out fifth is never touched. **There is nothing left for you to get wrong by hand — which is the entire reason Pipeline exists.**

Summary — data cleaning in twelve lines

1. Deletion is exponential in the number of columns. 5% of cells missing across 30 features left 114 of 569 patients.
2. Mean imputation preserves the mean and multiplies the variance by exactly m/n : measured 0.9486 against $\sqrt{0.9} = 0.9487$, and 0.5447 against $\sqrt{0.3} = 0.5477$.
3. That shrinkage propagates: $\text{corr}(\text{BMI}, y)$ fell from 0.5865 to 0.4538 at 40% imputed.
4. **MCAR** bias +0.12, **MAR** +1.35, **MNAR** -2.95 kg/m². MICE fixed MAR (+1.35 $\rightarrow$ -0.16) and could not fix MNAR (-2.95 $\rightarrow$ -2.58).
5. Under MNAR, mean imputation flipped a coefficient's sign: 5.60 $\rightarrow$ -0.41.
6. Dropping incomplete rows cost 0.05 of R^2 ; the choice among six imputers cost 0.003. **Spend your attention accordingly.**
7. `add_indicator=True` was worth +0.1675 accuracy when the missingness itself carried the signal.
8. **Thirty good readings and five stuck at 100: the $|z| > 3$ rule flagged none of them.** The modified z flagged exactly five.
9. Masking gets worse with more contamination: caught at $z = 5.38$ with one outlier, 3.11 with three, never from four onwards.
10. $|z|$ can never exceed $(n-1)/\sqrt{n}$, so below $n = 11$ the rule cannot fire at all.
11. On a skewed column the three rules flagged 6, 65 and 81 rows; after one Yeo-Johnson, 0, 1 and 0. Box-Cox took skewness 5.43 $\rightarrow$ 0.06 at $\lambda = -0.4357$.
12. Scaling before the split was worth +0.0034; deleting badly-fitted rows before the split was worth +0.1702. **Use a Pipeline and neither can happen.**

Before the next lecture

Read

notes.pdf — the full variance-shrinkage derivation, the $(n-1)/\sqrt{n}$ proof, and **10 practice questions with worked solutions**.

Do

Questions 2, 5 and 9. Question 5 is the three outlier rules by hand; question 9 is a debugging story you will meet for real.

Next: feature engineering

Scaling, encoding, and reducing the number of columns. Carry one thing across: **every one of those steps also learns a number from the training set**.

And a habit to start now

Before you fit anything: how many holes, why, and what does the histogram of each column look like?

Materials: <https://github.com/tali7c/Applied-Machine-Learning>

Sources: Han, Kamber and Pei, *Data Mining: Concepts and Techniques* (3e), Ch. 3.; Müller and Guido, *Introduction to Machine Learning with Python*, Ch. 3–4.; James, Witten, Hastie, Tibshirani and Taylor, *An Introduction to Statistical Learning with Python*, Ch. 5–6, 12.; Scikit-learn documentation (preprocessing, model selection), accessed 2026-08-04.